Saying "Not good, do it again" to AI causing huge energy problems

Edge User Input

GPU

2. Receives data and instructions from terminals and cloud.

1. Processes and caches data or low-latency response.

Giving AI an order for generating some initial ideas

Computing

Text encoding → Semantic parsing → Model inference → Generating Outcomes

Re-analyze the problem.

Asking AI for doing the task once again without any commnets

Not good, do it again

While this behavior is Low frequency but long-term, consumption accumulates over time through unconscious repetition.

Examples

When teachers make PPTs, students write essays, and in similar situations, they directly ask AI to generate content without clear ideas and iterate

Visible part

User → Easy access to AI

Ideas generating

Unconsciously

AI computing → Outcome → USer

Users repeatedly use AI without clear improvement goals, simply ask for redo

Repetitive behavior accumulates energy consumption.

Energy Loss

Invisible part

Planetary Consequence

Heat Cooling

Redundant computation Generating excessive heating.

Transmission power → Generating Heat (output) → Chip computation

All hardware requires cooling during operation. The water is consumed during process

GPU

Server chips

Hardware Requiring water resources (input)

Heat from repeated computing causes material loss in chips.

Repeated computation also damages the power supply module.

Repeated computing wasting more cooling water.

Heat and material loss accumulate continuously. Long-term thermal stress accelerates aging and shortens hardware lifespan.

1 AI image: 1080 J
5000-word paper: 90000 J
1 PPT: 14400 J

Image: 10800 J
Paper: 900000 J
PPT: 144000 J

Repeating 1000 computations:
Equivalent chip lifetime loss: ≈ 0.01%

Per 1,000,000 J energy consumption:
Carbon emissions: ≈ 0.06 kg
Cooling water consumption: ≈ 0.02 L

Material loss from heat generated during cooling and computation accumulates under repeated calculations.

Materials

Material extraction and cooling both require electricity.

Electricity

All the process requires electricity

Power consumption accumulates during repeated computations.

Computing, cooling and energy extraction consume electricity, with power accumulating in repeated operations.

Long-term repeated inference increases data center power load and baseline consumption.

Power use in transmission, storage accumulates with usage frequency.

Data repeating transmission also use electricity

User → Transmission loss → Computing power → Cooling power → Power supply → Long term use → Asking for redo

Power consumption accumulates linearly with operation count

Repeat 10 times:
Image: 3 Wh
Paper: 250 Wh

Repeat 100 times:
Image: 30 Wh
Paper: 2500 Wh

From quantitative to qualitative change